

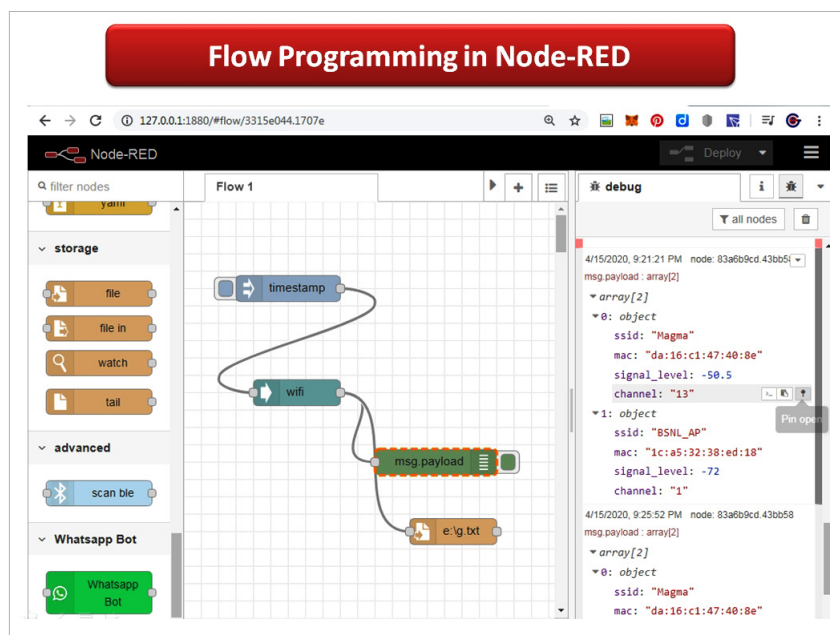


Figure 6: Creating a flow diagram in Node-RED with Wi-Fi interfacing

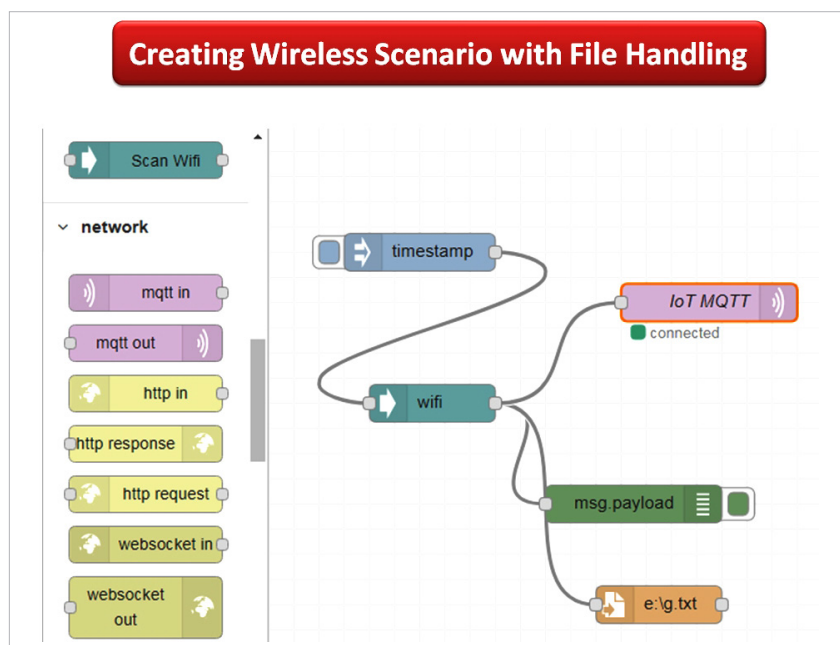


Figure 7: Creating an IoT scenario for real-time interfacing with MQTT broker

using the left palette of icons. Here, all the packages are visible and can be dragged across the dashboard.

The message queueing telemetry transport (MQTT) protocol is available in the Node-RED palette to help it work with IoT. It is used as an IoT broker for real-time interfacing and communication with sensors. An IoT scenario is created by the file handling module, capturing data from sensors and using the MQTT protocol to interface it with Wi-Fi. After running this scenario on a single click, the results are visible on the right side panel of Node-RED. Similar other structures can be created and deployed for dynamic communication with IoT devices.

Researchers and engineers can create IoT scenarios as per their requirements with the customisation and addition of packages on Node-RED.

Node-RED can interface with Raspberry Pi or Arduino to provide real-time communication with sensors, which can be used for smart agriculture based applications so that farmers can keep track of their fields and crops. Weather forecasting and climate analysis can also be done using such programming platforms. **END** 🐧

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